

MECHANICS: MOTION

Beginner-friendly WASSCE Physics Study Notes

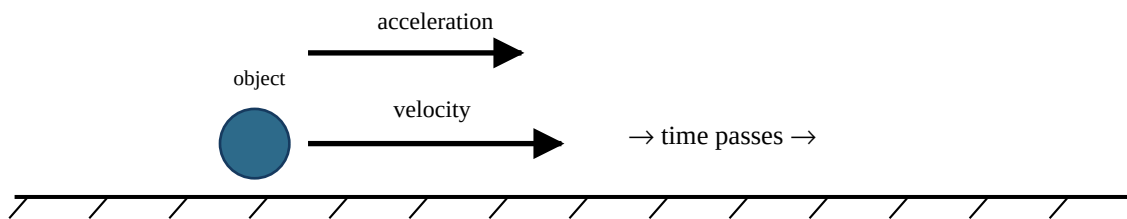
Clear explanations • Step-by-step derivations • Worked examples • Diagrams

How to use these notes

Do not memorize the equations first. Follow the reasoning that creates each equation. After each derivation, try the worked example without looking at the solution. The goal is understanding: know what each symbol means, why the formula works, and when to use it.

1. What Is Motion?

An object is said to be **in motion** when its position changes with time relative to a chosen reference point. For example, a car moving along a road changes its position relative to a roadside tree.



A useful question is: “Position compared with what?” Motion is therefore described relative to a reference point or frame of reference.

2. Distance and Displacement

Distance is the total length of the path travelled. It is a scalar quantity, so it has magnitude but no direction.

Displacement is the change in position from the starting point to the final point. It is a vector quantity, so it has both magnitude and direction.

Simple picture

If you walk 100 m east and then 60 m west, your distance is 160 m. Your displacement is 40 m east because your final position is only 40 m east of where you started.

$$\text{Displacement} = \text{final position} - \text{initial position}$$

The direction matters. A displacement can be positive, negative, or zero depending on the chosen positive direction.

3. Speed and Velocity

Speed describes how quickly distance is covered. **Velocity** describes how quickly displacement changes and includes direction.

$$\text{speed} = \text{distance} \div \text{time}$$

SI unit: m/s. Speed is a scalar quantity.

$$\text{velocity} = \text{displacement} \div \text{time}$$

SI unit: m/s. Velocity is a vector quantity.

Example: A runner completes one 400 m lap and returns to the starting point in 80 s. The average speed is $400/80 = 5 \text{ m/s}$. But the average velocity is zero because the final displacement is zero.

4. Acceleration

Acceleration measures how quickly velocity changes with time. A change can mean speeding up, slowing down, or changing direction.

$$a = (v - u) \div t$$

a = acceleration, u = initial velocity, v = final velocity, t = time.

The equation is not a magic rule. It comes directly from the meaning of acceleration:

Step 1: Change in velocity = final velocity – initial velocity.

Step 2: Acceleration = change in velocity $\div$ time.

Step 3: Therefore, $a = (v - u)/t$.

Step 4: Multiply both sides by t: $at = v - u$.

Step 5: Add u to both sides: **$v = u + at$** .

$$v = u + at$$

This is the first equation of uniformly accelerated motion. It applies when acceleration remains constant.

5. First Equation of Motion — A Full Derivation

We can now derive the first equation slowly, using only the definition of acceleration.

Start with: $a = (v - u)/t$

Multiply both sides by t : $at = v - u$

Add u to both sides: $u + at = v$

Rearrange: **$v = u + at$**

Why this matters

Every step is an ordinary algebraic operation. The equation is therefore not something you need to accept blindly: it follows from the definition of acceleration, provided acceleration is constant.

6. Second Equation of Motion — Derivation

For constant acceleration, velocity increases uniformly from u to v . The average velocity is therefore the average of the initial and final velocities:

$$\text{average velocity} = (u + v) \div 2$$

Because the velocity changes uniformly, the average velocity lies halfway between u and v .

Displacement equals average velocity multiplied by time:

$$s = [(u + v)/2]t$$

From the first equation, $v = u + at$. Substitute this for v :

$$s = [(u + (u + at))/2]t$$

Simplify inside the brackets:

$$s = [(2u + at)/2]t$$

Split the fraction:

$$s = [u + (1/2)at]t$$

Multiply through by t :

$$\mathbf{s = ut + (1/2)at^2}$$

$$\mathbf{s = ut + \frac{1}{2}at^2}$$

s is displacement, u is initial velocity, a is constant acceleration and t is time.

7. Third Equation of Motion — Derivation

We want an equation that connects velocity, acceleration and displacement without using time.

Start with the first equation:

$$v = u + at$$

Rearrange for time:

$$t = (v - u)/a$$

Now use the second equation:

$$s = [(u + v)/2]t$$

Substitute $t = (v - u)/a$:

$$s = [(u + v)/2] \times [(v - u)/a]$$

Multiply both sides by $2a$:

$$2as = (u + v)(v - u)$$

Use the difference-of-squares identity: $(A + B)(A - B) = A^2 - B^2$.

Therefore:

$$2as = v^2 - u^2$$

Rearrange:

$$v^2 = u^2 + 2as$$

$$v^2 = u^2 + 2as$$

Use this when time is not given or is inconvenient to use.

8. The Three Equations Together

$$v = u + at$$

Connects velocity, acceleration and time.

$$s = ut + \frac{1}{2}at^2$$

Connects displacement, initial velocity, acceleration and time.

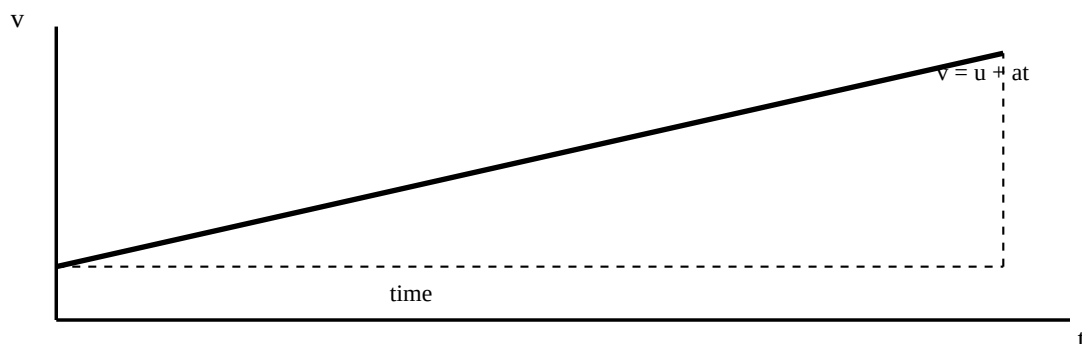
$$v^2 = u^2 + 2as$$

Connects velocity, acceleration and displacement without time.

9. Understanding Motion Graphs

Graphs are not just pictures. Their slopes and areas can represent physical quantities.

Velocity–time graph: the gradient represents acceleration.



The gradient is rise/run. On a velocity–time graph, rise is change in velocity and run is time. Therefore $\text{gradient} = \text{change in velocity} \div \text{time} = \text{acceleration}$.

Area under a velocity–time graph: for constant velocity, displacement = velocity $\times$ time, which is the rectangle's area. For changing velocity, the same idea extends to the area under the graph.

10. Free Fall

When an object falls near Earth's surface and air resistance is neglected, it accelerates downward at approximately **9.8 m/s²** (often rounded to 10 m/s² in school calculations). This acceleration is called acceleration due to gravity, g.

Sign convention

Choose a positive direction before solving. If upward is positive, downward gravitational acceleration is $-g$. If downward is positive, then g is positive. The mathematics must match your chosen direction.

11. Worked Example 1

Question

A car starts from rest and accelerates uniformly at 3 m/s² for 6 s. Find its final velocity.

Given: $u = 0 \text{ m/s}$, $a = 3 \text{ m/s}^2$, $t = 6 \text{ s}$

Required: v

Choose: $v = u + at$

Substitute: $v = 0 + (3)(6)$

Answer: $v = 18 \text{ m/s}$

Why? The car gains 3 m/s of velocity every second for 6 seconds, so its velocity increases by 18 m/s from its starting value of zero.

12. Worked Example 2

Question

A stone moves with an initial velocity of 4 m/s and accelerates uniformly at 2 m/s² for 5 s. Find its displacement.

Given: $u = 4 \text{ m/s}$, $a = 2 \text{ m/s}^2$, $t = 5 \text{ s}$

Required: s

Choose: $s = ut + \frac{1}{2}at^2$

Substitute: $s = (4)(5) + \frac{1}{2}(2)(5^2)$

$s = 20 + 25$

Answer: $s = 45 \text{ m}$

13. Common Beginner Mistakes

- Confusing distance with displacement.
- Confusing speed with velocity.
- Using the wrong equation simply because it looks familiar.
- Forgetting to convert units before substituting.
- Mixing signs after choosing a positive direction.
- Using the constant-acceleration equations when acceleration is not constant.
- Memorizing formulas without understanding what each symbol represents.

14. Quick Understanding Check

- Explain in your own words why a person can have non-zero distance but zero displacement.
- A cyclist changes velocity from 5 m/s to 17 m/s in 4 s. What is the acceleration?
- A body starts from rest and accelerates at 2 m/s² for 8 s. Find its final velocity.
- Explain why the gradient of a velocity–time graph represents acceleration.
- Derive $v = u + at$ starting from the definition of acceleration.
- Explain why the third equation is useful when time is not known.

Key idea to remember

Physics becomes easier when you understand where equations come from. Do not ask only, "Which formula should I use?" Ask, "What does this situation mean physically, and which relationship describes it?"

Prepared for the Hassan A Hakim Education & Career Resource Platform

Mechanics • Motion • Beginner → WASSCE foundation